

Ambrish.S

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EDUCATION

CHENNAI INSTITUTE OF TECHNOLOGY (2023-2027)

Bachelor of Technology (B.Tech) - Artificial Intelligence and Data Science

Cumulative Grade Point Average (CGPA) - 7.97/ 10

PROJECTS

Smart Attendance System with Facial Recognition Using RFID Technology:-

Tech Stack:-"Python, OpenCV, Haar Cascade, LBPH, FaceNet, Numpy, TensorFlow"

Built a hybrid attendance system combining facial recognition with RFID verification for accurate, tamper-resistant attendance tracking. Used Haar Cascade and LBPH/FaceNet for face detection and recognition, with a TensorFlow pipeline for model training and inference.

3D Avatar-based Indian Sign Language (ISL) Interpreter using Machine Language :-

Tech Stack:-"React.js, Node.js, Express.js, MongoDB, Three.js, WebGL, ML, Bootstrap 5"

Developed a web app that translates speech/text to Indian Sign Language using 3D avatars. Built with React.js, Three.js, and Node.js, it features real-time translation, interactive learning modules, and a responsive interface. The project enhances digital accessibility for the deaf community through intuitive 3D animations and user-friendly design.

Multimodal Edge AI System for Visual Quality Inspection with RAG-Powered Root Cause Analysis.

Tech Stack:-"OpenCV, Edge AI, RAG-Vector DB, LLM Reasoning, ML Pipeline, Frontend+API"

Designed an edge-deployed visual inspection system that detects quality defects in real time using a lightweight computer vision pipeline. Integrated a RAG-powered LLM reasoning layer over a vector database to automatically generate root cause analysis for detected defects.

PROFESSIONAL EXPERIENCE

Ogrelix:- (Web Development Intern)

June 2024-July 2024

Full-stack developer with hands-on experience building a centralized classroom management platform.

Skilled across both front-end and back-end development, with a focus on practical, usable solutions.

Chennai Institute of Technology:- (Research Intern)

May 2025-June 2025

Authored a review paper on neural network-based intrusion detection systems, evaluating CNN, RNN, and GNN architectures for enterprise network security. Examined real-time deployment challenges, adversarial robustness, and explainable AI to outline scalable, resilient IDS design approaches.

Thanthi TV :-(Election Data Operations)

May 2026

Handled live election data entry and real-time result updates during election coverage. Monitored incoming election result data for accuracy, consistency, and timely broadcasting support. Supported software-based election data handling processes for real-time reporting and analysis.

Shibaura Institute of Technology :-

July 2026-Aug 2026

Designed 15 U-Net variants for sparse-view photoacoustic tomography. Developed a novel hybrid attention module matching 94% of CBAM's SSIM gain with a 95% parameter reduction. Built a Radon-based simulation pipeline and a 100% accurate CNN classifier for adaptive reconstruction.

SKILLS AND CERTIFICATIONS

Programming Languages: Python, Java, JavaScript, C++, C, PHP, Swift, HTML5, CSS3

Frontend Development: React, Next.js, Tailwind CSS, AngularJS, Three.js, Vite

Backend & Frameworks: Node.js, NestJS, .NET

Databases: MongoDB, MySQL, PostgreSQL, Supabase, SQLite, Firebase

Cloud & DevOps: AWS, Google Cloud, Azure, Vercel, Jenkins, Git, GitHub

AI/ML & Data Science: TensorFlow, PyTorch, OpenCV, Pandas, NumPy, Matplotlib

1. NPTEL Cloud Computing

Oct 2024

2. NPTEL Reinforcement Learning

Nov 2025

3. CCNA: Enterprise Networking, Security & Automation

2025

4. Cisco: Python Essentials 1 & 2

May 2026

5. Cisco: Introduction to Modern AI

Jun 2026

6. Cisco: CyberOps Associate

Jun 2026

ACHIEVEMENTS & LANGUAGES

1.)TCS CODE VITA SEASON 13 Cleared ROUND 1 & ROUND 2 AIR(5303)

2.)Rajasthan Digifest x TiE Global Hackathon (Finalist)

3.)Google Big Code 2026:-Cleared Round 1 (TOP 1000 STUDENTS)

• **English** – Proficient

• **Kannada** – Native

• **Hindi** – Intermediate

• **Japanese (JLPT N4)** – Beginner (Currently learning)

CONTRIBUTIONS

• **Minaliya Oils**- Independently developed and deployed a responsive corporate website for Minaliya using Next.js. Implemented modern UI, intuitive navigation, and mobile responsiveness while translating business requirements into functional web pages and managing the project end-to-end. (<https://www.minaliya.com>)

• **Voko Run Club**:- Developed a dynamic website for a Chennai-based Run Club using React.js, building reusable React components and implementing smooth scroll-based transitions. Focused on creating an engaging, responsive, and interactive user experience with modern frontend design. (<https://www.vokoclub.in>)

VOLUNTEERING

• **Yuvenza Youth Club**:- Coordinated and managed 2000+ student events, contributing to event planning and execution (2024)

• **Takshashila Cultural Fest**:- Assisted in sponsorship coordination, outreach, and sponsor management for the cultural fest (2024–2025)

• **Talos Technical Symposium**:- Contributed to planning and organizing technical events, coordinating 1000+ participants and ensuring smooth event execution (2025)